MILITARY SPECIFICATION SHEET

# CAPACITOR, FIXED, ELECTROLYTIC (TANTALUM), CHIP, ESTABLISHED RELIABILITY, STYLES CWR13 AND CWR14

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-55365.

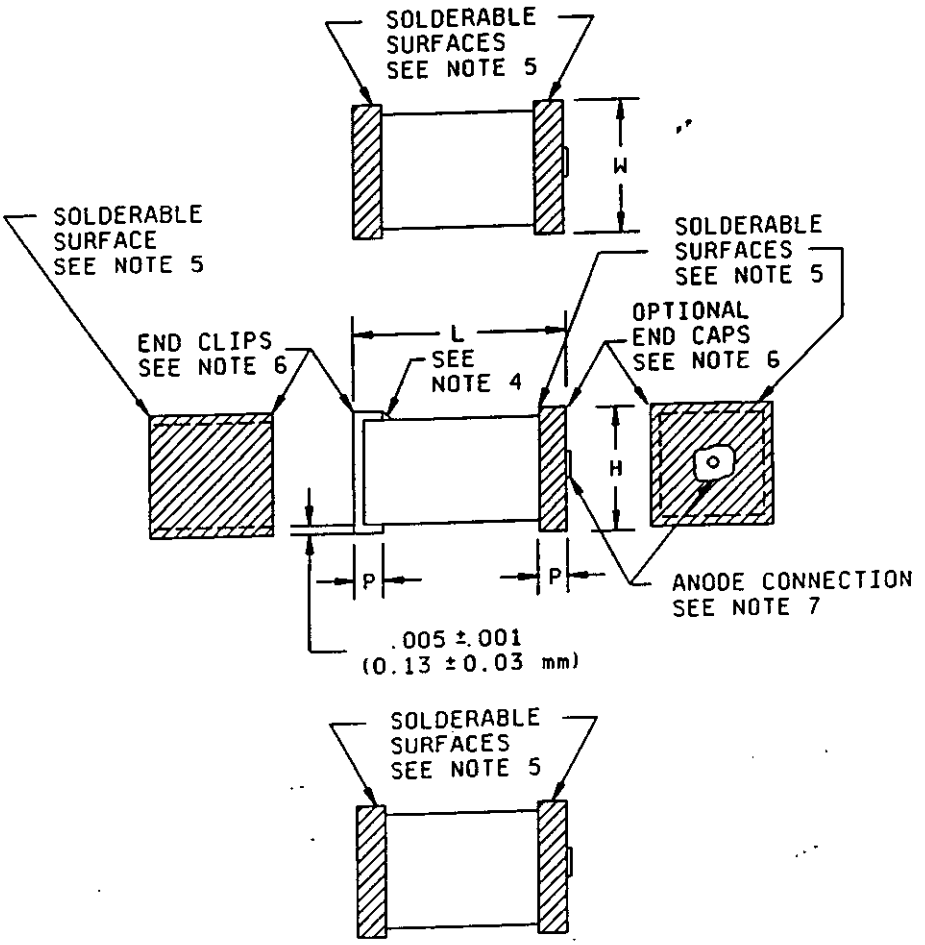



FIGURE 1. Style CWR13 (conformally coated) capacitors.



FSC 5910

AMSC N/A

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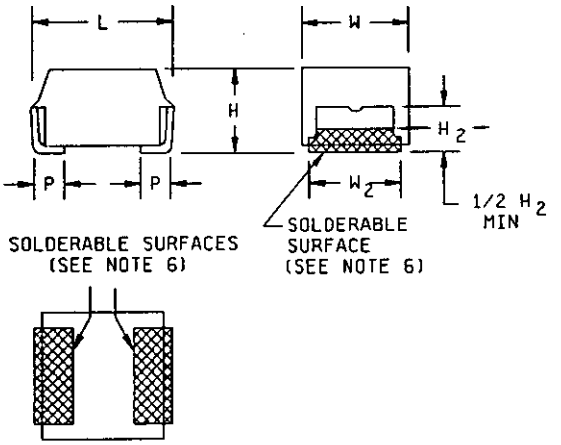
| Case size | L ±.015 (0.38) | W ±.015 (0.38) | H ±.015 (0.38) | P ±.005 (0.13) |
|---|---|---|---|---|
| A | .100 (2.54) | .050 (1.27) | .050 (1.27) | .030 (0.76) |
| B | .150 (3.81) | .050 (1.27) | .050 (1.27) | .030 (0.76) |
| C | .200 (5.08) | .050 (1.27) | .050 (1.27) | .030 (0.76) |
| D | .150 (3.81) | .100 (2.54) | .050 (1.27) | .030 (0.76) |
| E | .200 (5.08) | .100 (2.54) | .050 (1.27) | .030 (0.76) |
| F | .220 (5.59) | .135 (3.43) | .070 (1.78) | .030 (0.76) |
| G | .265 (6.73) | .110 (2.79) | .110 (2.79) | .050 (1.27) |
| H | .285 (7.24) | .150 (3.81) | .110 (2.79) | .050 (1.27) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
4. Cathode terminal may be epoxy coated on one surface for polarity identification.
5. Solderable surfaces are only those surfaces designated as such.
6. Anode and cathode terminals may be two-sided through five-sided, and may not be the same on some designs, but shall meet all specification requirements.
7. The anode terminal shall be identified by the riser wire connection, which may extend the case size .015 inch (0.38 mm) maximum. The riser wire weld area is not solderable. The riser wire connection shall have a continuous weld of a minimum of 25 percent of the contact surface.
8. When solder coated terminations are required, add .015 inch (0.38 mm) to termination dimension tolerances.

FIGURE 1. Style CWR13 (conformally coated) capacitors - Continued.



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| Case size | L ±.015 (0.38) | W ±.015 (0.38) | H ±.015 (0.38) | P ±.005 (0.13) | $W_2$ ±.005 (0.13) | $H_2$ Min |
|---|---|---|---|---|---|---|
| A | .100 (2.54) | .050 (1.27) | .050 (1.27) | .030 (0.76) | .050 (1.27) | .030 (0.76) |
| B | .150 (3.81) | .050 (1.27) | .050 (1.27) | .030 (0.76) | .050 (1.27) | .030 (0.76) |
| C | .200 (5.08) | .050 (1.27) | .050 (1.27) | .030 (0.76) | .050 (1.27) | .030 (0.76) |
| D | .150 (3.81) | .100 (2.54) | .050 (1.27) | .030 (0.76) | .095 (2.41) | .030 (0.76) |
| E | .200 (5.08) | .100 (2.54) | .050 (1.27) | .030 (0.76) | .095 (2.41) | .030 (0.76) |
| F | .220 (5.59) | .135 (3.43) | .070 (1.78) | .030 (0.76) | .130 (3.30) | .040 (1.02) |
| G | .265 (6.73) | .110 (2.79) | .110 (2.79) | .050 (1.27) | .105 (2.67) | .060 (1.52) |
| H | .285 (7.24) | .150 (3.81) | .110 (2.79) | .050 (1.27) | .145 (3.68) | .060 (1.52) |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. These capacitors are designed for mounting by dip soldering, thermo-compression bonding, reflow soldering, or other conventional means.
4. The anode (+) terminal shall be identified by a marking on the case.
5. The termination width ($W_2$) shall be maintained to the top of the solderable area.
6. Solderable surfaces are only those surfaces designated as such. Termination edges are not considered solderable.
7. When solder coated terminations are required, add .015 inch (0.38 mm) to the termination dimension tolerances.

FIGURE 2. Style CWR14 molded chip.



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**REQUIREMENTS:**

Design and construction:

Dimensions and configuration: See figures 1 and 2.

Case type: Chip, conformally coated or molded.

Material: Sintered tantalum.

Leads: Unleaded.

Termination finish: In accordance with termination finishes B, C, H, or K of MIL-C-55365.

DC rated voltage: See table I. Above 85°C, voltage derating is required in accordance with MIL-C-55365.

Operating temperature range: -55°C to +125°C.

Failure rate level: In accordance with MIL-C-55365.

DC leakage (DCL): See table I.

Capacitance (Cap.): See table I.

Cap. tolerance: (J) ±5 percent, (K) ±10 percent, or (M) ±20 percent.

Dissipation factor (DF): See table I.

Resistance to soldering heat: In accordance with MIL-C-55365.

Stability at low and high temperatures: In accordance with MIL-C-55365.

Step 1 (+25°C):

DCL: See table I.

Cap.: Shall not exceed tolerance of table I value.

DF: See table I.

Step 2 (-55°C):

ΔCap.: Shall not exceed ±10 percent of initial measurement.

DF: See table I.

Step 3 (+25°C):

DCL: See table I.

ΔCap.: Shall not exceed ±5 percent of initial +25°C measurement.

DF: See table I.

Step 4 (+85°C):

DCL: See table I.

ΔCap.: Shall not exceed ±10 percent of initial +25°C measurement.

DF: See table I.



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TABLE I. Styles CWR13 and CWR14 characteristics.

| PIN 1/ 2/ | DC rated voltage (85°C) | Cap. (nom) | DC leakage (max) | | | Dissipation factor (max) | | | ESR max 100 kHz | Case size |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | +25°C | +85°C | +125°C | +25°C | +85°C +125°C | -55°C | +25°C | |
| | Volts | µF | µA | µA | µA | % | % | % | Ohms | |
| CWR1-C-685-- | 4 | 6.8 | 0.5 | 5 | 6 | 6 | 8 | 8 | 3.5 | A |
| CWR1-C-106-- | 4 | 10.0 | 0.5 | 5 | 6 | 8 | 10 | 12 | 2.0 | B |
| CWR1-C-156-- | 4 | 15.0 | 0.6 | 6 | 7.2 | 8 | 10 | 12 | 1.5 | C |
| CWR1-C-226-- | 4 | 22.0 | 0.9 | 9 | 10.8 | 8 | 10 | 12 | 1.0 | D |
| CWR1-C-336-- | 4 | 33.0 | 1.4 | 14 | 16.8 | 8 | 10 | 12 | 1.0 | E |
| CWR1-C-107-- | 4 | 100.0 | 4 | 40 | 48 | 10 | 12 | 12 | 0.5 | F |
| CWR1-C-157-- | 4 | 150.0 | 6 | 60 | 72 | 10 | 12 | 12 | 0.5 | G |
| CWR1-C-227-- | 4 | 220.0 | 9 | 90 | 108 | 10 | 12 | 12 | 0.5 | H |
| CWR1-D-475-- | 6 | 4.7 | 0.5 | 5 | 6 | 6 | 8 | 8 | 3.5 | A |
| CWR1-D-685-- | 6 | 6.8 | 0.5 | 5 | 6 | 6 | 8 | 8 | 2.0 | B |
| CWR1-D-106-- | 6 | 10.0 | 0.6 | 6 | 7.2 | 8 | 10 | 12 | 1.5 | C |
| CWR1-D-156-- | 6 | 15.0 | 0.9 | 9 | 10.8 | 8 | 10 | 12 | 1.0 | D |
| CWR1-D-226-- | 6 | 22.0 | 1.4 | 14 | 16.8 | 8 | 10 | 12 | 1.0 | E |
| CWR1-D-686-- | 6 | 68.0 | 4 | 40 | 48 | 10 | 12 | 12 | 0.7 | F |
| CWR1-D-107-- | 6 | 100.0 | 8 | 80 | 96 | 10 | 12 | 12 | 0.5 | G |
| CWR1-D-157-- | 6 | 150.0 | 9 | 90 | 108 | 10 | 12 | 12 | 0.5 | H |
| CWR1-F-335-- | 10 | 3.3 | 0.5 | 5 | 6 | 6 | 8 | 8 | 3.5 | A |
| CWR1-F-475-- | 10 | 4.7 | 0.5 | 5 | 6 | 6 | 8 | 8 | 2.0 | B |
| CWR1-F-685-- | 10 | 6.8 | 0.7 | 7 | 8.4 | 6 | 8 | 8 | 2.0 | C |
| CWR1-F-106-- | 10 | 10.0 | 1 | 10 | 12 | 8 | 10 | 10 | 1.5 | D |
| CWR1-F-156-- | 10 | 15.0 | 1.5 | 15 | 18 | 8 | 10 | 10 | 1.0 | E |
| CWR1-F-476-- | 10 | 47.0 | 4.7 | 47 | 56.4 | 10 | 12 | 12 | 0.7 | F |
| CWR1-F-686-- | 10 | 68.0 | 7 | 70 | 84 | 10 | 12 | 12 | 0.7 | G |
| CWR1-F-107-- | 10 | 100.0 | 10 | 100 | 120 | 10 | 12 | 12 | 0.5 | H |
| CWR1-H-225-- | 15 | 2.2 | 0.5 | 5 | 6 | 6 | 8 | 8 | 4.0 | A |
| CWR1-H-335-- | 15 | 3.3 | 0.5 | 5 | 6 | 6 | 8 | 8 | 2.5 | B |
| CWR1-H-475-- | 15 | 4.7 | 0.8 | 8 | 9.6 | 6 | 8 | 8 | 2.5 | C |
| CWR1-H-685-- | 15 | 6.8 | 1 | 10 | 12 | 6 | 8 | 8 | 2.0 | D |
| CWR1-H-106-- | 15 | 10.0 | 1.6 | 16 | 19.2 | 6 | 8 | 8 | 1.5 | E |
| CWR1-H-336-- | 15 | 33.0 | 5 | 50 | 60 | 8 | 8 | 10 | 0.8 | F |
| CWR1-H-476-- | 15 | 47.0 | 8 | 80 | 96 | 8 | 8 | 10 | 0.8 | G |
| CWR1-H-686-- | 15 | 68.0 | 11 | 110 | 132 | 8 | 8 | 10 | 0.8 | H |
| CWR1-J-155-- | 20 | 1.5 | 0.5 | 5 | 6 | 6 | 8 | 8 | 6.0 | A |
| CWR1-J-225-- | 20 | 2.2 | 0.5 | 5 | 6 | 6 | 8 | 8 | 3.5 | B |
| CWR1-J-335-- | 20 | 3.3 | 0.8 | 8 | 9.6 | 6 | 8 | 8 | 3.0 | C |
| CWR1-J-475-- | 20 | 4.7 | 1 | 10 | 12 | 6 | 8 | 8 | 2.5 | D |
| CWR1-J-685-- | 20 | 6.8 | 1.4 | 14 | 16.8 | 6 | 8 | 8 | 2.0 | E |
| CWR1-J-226-- | 20 | 22.0 | 4 | 40 | 48 | 6 | 8 | 8 | 1.0 | F |
| CWR1-J-336-- | 20 | 33.0 | 7 | 70 | 84 | 6 | 8 | 8 | 1.0 | G |
| CWR1-J-476-- | 20 | 47.0 | 10 | 100 | 120 | 6 | 8 | 8 | 1.0 | H |
| CWR1-K-105-- | 25 | 1.0 | 0.5 | 5 | 6 | 6 | 8 | 8 | 8.0 | A |
| CWR1-K-155-- | 25 | 1.5 | 0.5 | 5 | 6 | 6 | 8 | 8 | 5.0 | B |
| CWR1-K-225-- | 25 | 2.2 | 0.8 | 8 | 9.6 | 6 | 8 | 8 | 3.5 | C |
| CWR1-K-335-- | 25 | 3.3 | 0.9 | 9 | 10.8 | 6 | 8 | 8 | 3.5 | D |

See footnotes at end of table.



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TABLE I. Styles CWR13 and CWR14 characteristics - Continued.

| PIN 1/ 2/ | DC rated voltage (85°C) | Cap. (nom) | DC leakage (max) | | | Dissipation factor (max) | | | ESR max 100 kHz +25°C | Case size |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | +25°C | +85°C | +125°C | +25°C | +85°C +125°C | -55°C | | |
| | Volts | µF | µA | µA | µA | % | % | % | Ohms | |
| CWR1-K-475-- | 25 | 4.7 | 1 | 10 | 12 | 6 | 8 | 8 | 2.5 | E |
| CWR1-K-106-- | 25 | 10.0 | 2.5 | 25 | 30 | 6 | 8 | 8 | 2.0 | F |
| CWR1-K-156-- | 25 | 15.0 | 3.8 | 38 | 45.6 | 6 | 8 | 8 | 1.0 | F |
| CWR1-K-226-- | 25 | 22.0 | 6 | 60 | 72 | 6 | 8 | 8 | 1.0 | G |
| CWR1-K-336-- | 25 | 33.0 | 9 | 90 | 108 | 6 | 8 | 8 | 1.0 | H |
| CWR1-M-474-- | 35 | 0.47 | 0.5 | 5 | 6 | 6 | 8 | 8 | 12.0 | A |
| CWR1-M-105-- | 35 | 1.0 | 0.5 | 5 | 6 | 6 | 8 | 8 | 6.0 | B |
| CWR1-M-155-- | 35 | 1.5 | 0.5 | 5 | 6 | 6 | 8 | 8 | 5.0 | C |
| CWR1-M-225-- | 35 | 2.2 | 0.8 | 8 | 9.6 | 6 | 8 | 8 | 3.5 | D |
| CWR1-M-335-- | 35 | 3.3 | 1 | 10 | 12 | 6 | 8 | 8 | 2.5 | E |
| CWR1-M-475-- | 35 | 4.7 | 1.7 | 17 | 20.4 | 6 | 8 | 8 | 2.5 | F |
| CWR1-M-685-- | 35 | 6.8 | 2.4 | 24 | 28.8 | 6 | 8 | 8 | 2.0 | F |
| CWR1-M-106-- | 35 | 10.0 | 4 | 40 | 48 | 6 | 8 | 8 | 2.0 | G |
| CWR1-M-156-- | 35 | 15.0 | 6 | 60 | 72 | 6 | 8 | 8 | 2.0 | H |
| CWR1-N-104-- | 50 | 0.10 | 0.5 | 5 | 6 | 6 | 8 | 8 | 30.0 | A |
| CWR1-N-154-- | 50 | 0.15 | 0.5 | 5 | 6 | 6 | 8 | 8 | 20.0 | A |
| CWR1-N-224-- | 50 | 0.22 | 0.5 | 5 | 6 | 6 | 8 | 8 | 15.0 | B |
| CWR1-N-334-- | 50 | 0.33 | 0.5 | 5 | 6 | 6 | 8 | 8 | 15.0 | B |
| CWR1-N-474-- | 50 | 0.47 | 0.5 | 5 | 6 | 6 | 8 | 8 | 10.0 | C |
| CWR1-N-684-- | 50 | 0.68 | 0.5 | 5 | 6 | 6 | 8 | 8 | 8.0 | D |
| CWR1-N-105-- | 50 | 1.0 | 0.5 | 5 | 6 | 6 | 8 | 8 | 5.0 | E |
| CWR1-N-155-- | 50 | 1.5 | 0.5 | 5 | 6 | 6 | 8 | 8 | 4.0 | F |
| CWR1-N-225-- | 50 | 2.2 | 1.1 | 11 | 13.2 | 6 | 8 | 8 | 3.5 | F |
| CWR1-N-335-- | 50 | 3.3 | 2.0 | 20 | 24 | 6 | 8 | 8 | 2.5 | G |
| CWR1-N-475-- | 50 | 4.7 | 3.0 | 30 | 36 | 6 | 8 | 8 | 1.0 | H |

1/ Complete PIN shall include additional symbols to indicate style, termination finish, capacitance tolerance, and failure rate level.

2/ Styles CWR13 and CWR14 are, with the acquiring agency approval, considered to be interchangeable, provided the size, capacitance, capacitance tolerance, failure rate level, and rated voltage remain the same.



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Step 5 (+125°C):

DCL: See table I.

ΔCap.: Shall not exceed ±15 percent of initial +25°C measurement.

DF: See table I.

Step 6 (+25°C):

DCL: See table I.

ΔCap.: Shall not exceed ±5 percent of initial +25°C measurement.

DF: See table I.

Surge voltage: In accordance with MIL-C-55365.

DCL: See table I.

ΔCap.: Within ±5 percent of initial value.

DF: See table I.

Life: Method 108 of MIL-STD-202.

2,000 hours:

At +25°C:

DCL: See table I.

ΔCap.: Shall not exceed ±5 percent of initial measured value.

DF: See table I.

At +85°C:

DCL: See table I.

At +125°C:

DCL: See table I.

10,000 hours: Exponential only.

At +25°C:

DCL: 200 percent of initial requirement.

ΔCap.: Shall not exceed ±10 percent of initial measured value.

DF: See table I.

At +85°C:

DCL: 200 percent of initial requirement.

At +125°C:

DCL: 200 percent of initial requirement.



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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - SH
Air Force - 17, 99
NASA - NA
DLA - ES

User activities:
Army - AR
Navy - AS, CG, MC, OS
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

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